

# Deepfake Detection Using CNN and Capsule Networks: A Comparative Analysis and Performance Evaluation

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**Abstract—** Deepfake technology has rapidly evolved with advancements in deep learning, making it increasingly difficult to distinguish between real and manipulated media. This paper presents an implementation-based approach for deepfake detection using Convolutional Neural Networks (CNN) and Capsule Networks.

The proposed system utilizes CNN for feature extraction from video frames, while Capsule Networks are used to preserve spatial relationships between facial features. The model is evaluated on benchmark datasets such as FaceForensics++ and DFDC. Performance is analyzed using metrics including accuracy, precision, recall, and F1-score.

Experimental observations indicate that CNN models effectively capture visual artifacts, while Capsule Networks improve the detection of structural inconsistencies. A hybrid CNN-Capsule approach is also explored to enhance detection performance and generalization across datasets. Although the hybrid model introduces additional computational complexity, it demonstrates improved robustness compared to standalone models.

This study highlights the importance of combining feature extraction and spatial relationship modeling for reliable deepfake detection in real-world applications.



**Keywords —** Deepfake detection, Convolutional Neural Network, Capsule Networks, Synthetic Media, Artificial Intelligence

## I. Introduction

Deepfake technology has advanced rapidly with the growth of artificial intelligence and deep learning, making it increasingly difficult to distinguish between real and manipulated media. While deepfakes have useful applications in media and entertainment, they also pose serious risks such as misinformation, identity theft, and privacy violations.

To address this problem, deep learning-based methods such as Convolutional Neural Networks (CNNs) are widely used for detecting visual inconsistencies in images and videos. However, CNNs often fail to preserve spatial relationships between facial features, which limits their performance on complex manipulations.

Capsule Networks have been introduced to overcome this limitation by maintaining spatial hierarchies and part-whole relationships within images. In this paper, we implement and analyze deepfake detection using CNN and Capsule Networks, and further explore a hybrid approach to improve performance. The proposed methods are evaluated using standard metrics such as accuracy, precision, recall, and F1-score.

## II. Background

With the rapid development of deep learning, the deepfake technique has been promoted in the deepfake community for the production of realistic-mimicking manipulated image and video. The deepfake media are produced by using some complex neural network structure to modify the identity, facial expression, and even the speech. Although such technology is beneficial for some applications such as entertainment, media production, and virtual environments, it also causes many issues such as misinformation, identity theft, trust, etc.

Researchers have investigated different approaches for automatically detecting deepfakes, which have been widely used based on their capability to easily obtain hierarchical characteristics of images, through the utilization of Convolutional Neural Networks. These techniques can also

be used to detect issues with deepfakes, including texture irregularities, blending artifacts and pixel-level distortions. However, as noted earlier, CNNs generally do poorly at identifying/characterizing detailed structures of spatial configuration and may not be able to generalize well to new datasets or emerging types of deepfake detection techniques. Forensic detection of tampered images is a popular topic. However, most detection methods are limited in terms of spatial hierarchies and the relationships between features. Capsule Networks have been proposed as a potential alternative which can efficiently preserve the spatial hierarchies and relationships between features to identify subtle manipulations. And they are also relatively efficient. Furthermore, many other detection methods look at the lightweight detection model for real-time detection or forensic-based detection which combine techniques together. But the performance could be improved in generalization, robustness, and computational efficiency. Hence, more and more recent research focuses on hybrid methods for combining CNN and capsule networks for better detection performance.

In addition, deepfake generation techniques have become increasingly sophisticated, making detection more challenging. Modern approaches require models that can analyze both low-level visual artifacts and high-level structural inconsistencies in facial features.

Furthermore, the rapid evolution of generative models such as Generative Adversarial Networks (GANs) has significantly improved the realism of deepfake content. These advancements make it difficult for traditional detection techniques to remain effective over time.

Therefore, there is a growing need for robust detection systems that can adapt to new manipulation techniques while maintaining high accuracy and reliability. This has led to the exploration of hybrid models that combine multiple deep learning approaches for improved performance.

### III. Literature Review

#### 3.1. Traditional methods

In the beginning, detection methods for deepfakes were based on specific visual cues created by hand (for example, facial landmarks, eye blinking patterns, lighting inconsistencies and compression artifacts) and were analyzed manually – making them more straightforward to perform without needing large amounts of data or computational resources.

Due to improvements in the technology used to create deep fakes, the visible artifacts created by this process have decreased attention towards their use for identifying deep fakes – meaning there are no longer clear signs of manipulation. Therefore, handcrafted methods of feature detection do not generalize well across different types of manipulated images/environments and fail to perform adequately in real life settings.

The commonly used deepfake detection methods can be summarized as follows:

- **Deep Learning Models:** These models provide high detection performance by learning complex visual patterns from data. However, they often suffer from limited robustness when applied to diverse and unseen deepfake types.
- **Capsule Networks:** Capsule Networks are designed to preserve spatial relationships between features, enabling better detection of structural inconsistencies. Despite their advantages, they require further evaluation across varied datasets.
- **Lightweight Detection Models:** These models focus on improving real-time detection performance and computational efficiency. However, this often comes at the cost of reduced accuracy compared to more complex models.
- **Forensic-based Platforms:** These approaches provide robust detection capabilities by analyzing digital artifacts and inconsistencies. However, they are computationally intensive and require significant resources.

#### 3.2. CNN-Based Methods

With the rise of deep learning, Convolutional Neural Networks (CNNs) have become one of the most widely used approaches for deepfake detection. Popular architectures such as Xception, ResNet, and EfficientNet have shown strong performance in identifying manipulated content. CNNs are capable of learning hierarchical features, including edges, textures, and subtle visual inconsistencies that may not be easily detected by traditional methods.

Despite their success, CNN-based models have certain limitations. Pooling operations can lead to the loss of important spatial information, which affects the model's ability to capture relationships between different facial features. This limitation reduces their effectiveness in detecting complex and unseen deepfake samples, especially when spatial relationships between facial features are important.

- **Deep Learning Models:** These models provide high detection performance by learning complex visual patterns

from data. However, they often suffer from limited robustness when applied to diverse and unseen deepfake

#### 3.3. Capsule Networks-Based Approaches

Capsule Networks (CapsNets) have been introduced as an alternative to traditional CNN architectures, with a focus on preserving spatial hierarchies within images. Unlike CNNs, CapsNets use groups of neurons, called capsules, to represent features along with their orientation and position. This allows the network to better understand the relationships between different parts of an image.

CapsNets are particularly effective in modeling part-whole relationships, which is important for detecting subtle manipulations in facial structures. Additionally, they can achieve competitive performance with fewer parameters compared to deep CNN models. However, this capability makes Capsule Networks suitable for detecting structural inconsistencies in deepfake images, which motivates their use in the proposed approach of this study.

### 3.4. CNN + Capsule Networks Approaches

Recent research has explored the combination of CNNs and Capsule Networks to leverage the strengths of both architectures. In such approaches, CNNs are used for efficient feature extraction, while Capsule Networks are applied to preserve spatial relationships and capture structural information within the image.

Hybrid approaches combine CNNs and Capsule Networks to utilize the strengths of both models. CNNs are used for extracting meaningful features, while Capsule Networks preserve spatial relationships between these features. This combination improves robustness and generalization, making it suitable for deepfake detection. In this work, a similar hybrid approach is implemented and analyzed to evaluate its performance.

## IV. Methodology

In this study, a deepfake detection system is implemented using Convolutional Neural Networks (CNN) and Capsule Networks. The overall workflow of the proposed system is illustrated in Fig. 1.

The process begins with input video data, from which frames are extracted for analysis. Face detection is then applied to each frame using standard techniques to isolate facial regions. This ensures that only relevant features are considered during analysis.

The detected faces are preprocessed by resizing and normalization to maintain consistency across the dataset. This step improves model performance and stability during training.

The preprocessed images are then passed through a CNN model, which is responsible for extracting hierarchical features such as textures, edges, and visual artifacts. These features capture important information related to possible manipulations in the image.

Following feature extraction, a Capsule Network is applied to preserve spatial relationships between facial components. This helps in identifying structural inconsistencies that may not be captured by CNNs alone.

Finally, the extracted features are classified as real or fake using a fully connected classification layer. The model is trained and evaluated using standard datasets, and performance is measured using metrics such as accuracy, precision, recall, and F1-score.

The combination of CNN and Capsule Networks allows the system to leverage both feature extraction and spatial relationship modeling, resulting in improved deepfake detection performance.

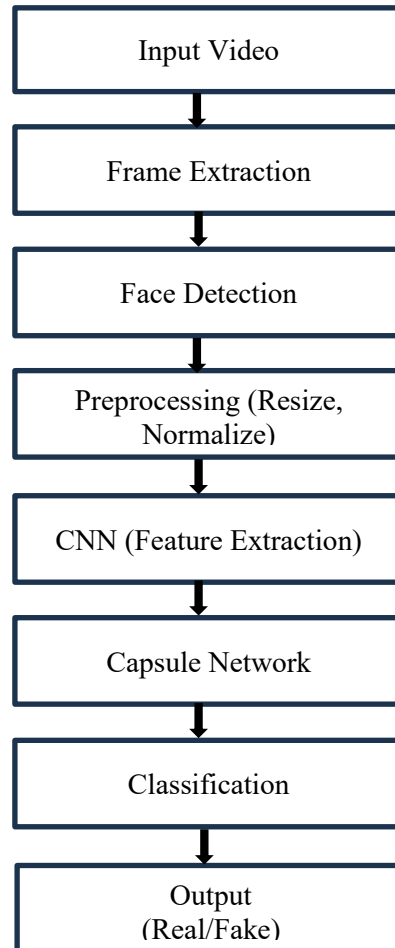


Fig. 1. Workflow of the proposed deepfake detection system

## V. Comparative Analysis

Deepfake detection methods show varying strengths and limitations. CNN models achieve high accuracy but lose spatial relationships. Capsule Networks preserve structure but are complex. The proposed hybrid CNN–Capsule model improves accuracy, robustness, and generalization across datasets. The comparison of different deepfake detection approaches is presented in Table 1.

Capsule Networks overcome this limitation by allowing for the retention of both part–whole and hierarchical relationships within an image, making them very useful for the detection of small manipulations. Rather than viewing Capsule Networks and Convolutional Neural Networks (CNNs) as two competing options, they can be viewed as two complementary options. CNNs are good at extracting meaningful features from images, while the use of Capsule Networks improves how those features relate to one another.

| S. No. | Year-Publication                 | Methods                                   | Advantage                                                                                             | Limitation/Future Work                       |
|--------|----------------------------------|-------------------------------------------|-------------------------------------------------------------------------------------------------------|----------------------------------------------|
| 1.     | Recent Studies                   | CNN + Capsule Network<br>(Proposed Model) | Combines feature extraction and spatial relationship modeling; improved robustness and generalization | Slight increase in computational complexity. |
| 2.     | 2025 – IEEE Access               | Dense-Swish-CNN + Bi-LSTM                 | High spatial-temporal accuracy                                                                        | High computational cost                      |
| 3.     | 2024 – Cyber Security            | Secure Deepfake Mitigation Framework      | Addresses societal and security issues                                                                | Limited technical evaluation                 |
| 4.     | 2024 – Procedia Computer Science | GAN-CNN Ensemble Model                    | Robust against catastrophic forgetting                                                                | Complex training pipeline                    |
| 5.     | 2024 – AI Review                 | Trend Analysis Survey                     | Comprehensive trend analysis                                                                          | No experimental validation                   |
| 6.     | 2024 - CVPR                      | Binary Neural networks                    | Real-time detection                                                                                   | Slight accuracy trade-off                    |
| 7.     | 2024 - CVPR                      | Multimodal Prototype-based Model          | High robustness & explainability                                                                      | Requires multimodal inputs                   |
| 8.     | 2024 - CVPR                      | CNN Up Sampling Analysis                  | Improved generalization                                                                               | Architecture                                 |
| 9.     | 2024 - Electronics               | Survey on Datasets & Algorithms           | Structured taxonomy                                                                                   | Lacks benchmarking                           |
| 10.    | 2024 - AAAI                      | Forgery Clue Extraction                   | Detects fine-grained artifacts                                                                        | High computation                             |
| 11.    | 2024 – IJAI Systems              | Ensemble Deep Learning                    | Improved reliability                                                                                  | Increased model size                         |

Table 1: Comparative analysis of deepfake detection methods

Traditional ways of finding out if a video is real or fake are not enough anymore because fake videos have become much more complicated. Using deep learning technologies like Convolutional Neural Network (CNN) has improved the ability of machine learning programs to find fake videos by helping them to notice complex visual patterns and inconsistencies in images. However, CNNs can create more problems than they solve when it comes to keeping the spatial relationships between different parts of the face.

Hybrid approaches that combine CNNs and Capsule Networks leverage the strengths of both models. This combination improves robustness and generalization, making it more suitable for detecting a wide range of deepfake manipulations. However, further research is needed to optimize these models for real-time performance and large-scale deployment.

## VI. Challenges

Despite significant progress in deepfake detection, several challenges remain that limit the effectiveness of current approaches in real-world scenarios.

1. Many models perform well on training datasets but fail to generalize to unseen or real-world data.
2. Deep learning models require large and diverse datasets, but existing datasets often do not reflect real-world conditions.
3. Advanced models, especially hybrid approaches, require high computational resources, limiting their use in real-time applications.
4. CNN-based models may lose spatial information, while Capsule Networks are computationally complex and not yet fully optimized.
5. Deepfake generation techniques are continuously improving, making detection increasingly challenging.
6. Many deep learning models lack interpretability, making it difficult to understand their decision-making process.

Overall, these challenges indicate that there is still room for improvement. Combining CNNs with Capsule Networks is a promising direction, but further work is required to improve efficiency and reliability in real-world applications.

## VII. Literature Findings

A review of the existing studies shows a clear progression in deepfake detection methods over time. Earlier approaches mainly relied on handcrafted features, which were relatively simple and easy to implement. However, as deepfake generation techniques improved, these methods became less reliable, particularly in real-world situations where manipulations are more refined and difficult to detect.

With the growth of deep learning, CNN-based methods became widely adopted due to their ability to learn detailed visual patterns from images and videos. These models have been effective in identifying subtle inconsistencies, but they are limited in capturing the relationships between different facial features. This limitation affects their performance when dealing with more complex or unseen manipulations.

To improve detection performance, hybrid approaches combining CNNs with temporal models such as RNNs and LSTMs were introduced. These methods are useful for analyzing video data, as they consider both spatial and temporal information. However, they often require higher

higher computational resources, which can restrict their use in practical applications.

Capsule Networks were later explored as an alternative approach, mainly because of their ability to preserve spatial relationships within images. This makes them suitable for detecting structural inconsistencies that may not be captured by CNNs alone. Although promising, their use is still limited due to challenges related to scalability and implementation.

Recent research indicates that combining CNNs with Capsule Networks can offer a more effective solution. CNNs help in extracting meaningful features, while Capsule Networks improve the understanding of how these features are related. This combination has shown better robustness and adaptability, suggesting its potential for handling a wide range of deepfake manipulations.

## VIII. Future Work

Future research should focus on developing efficient and scalable deepfake detection models that balance accuracy and computational cost. While the proposed hybrid CNN–Capsule approach improves detection performance, further optimization is required to reduce computational complexity for real-time applications.

In addition, future work can explore the use of larger and more diverse datasets to improve model generalization across different types of deepfake manipulations. This will help ensure that detection systems remain effective in real-world scenarios.

Another important direction is the integration of multimodal approaches, combining visual, audio, and temporal features for more comprehensive detection. Furthermore, improving model interpretability will help in understanding the decision-making process of deep learning models, increasing trust and reliability.

Overall, advancing hybrid architectures and incorporating adaptive learning techniques can further enhance the robustness and applicability of deepfake detection systems.

Moreover, future research can focus on developing lightweight and energy-efficient models that can be deployed on edge devices such as mobile phones and surveillance systems. This will enable real-time deepfake detection in practical environments where computational resources are limited.

In addition, future work can investigate the robustness of deepfake detection models against adversarial attacks and unseen manipulation techniques. As deepfake generation methods continue to evolve, it is essential to design adaptive systems that can continuously learn and update their detection strategies. This will ensure long-term reliability and effectiveness of deepfake detection systems in dynamic real-world environments.

## IX. Results and Analysis

The proposed deepfake detection model was evaluated using benchmark datasets such as FaceForensics++ and DFDC. The dataset was divided into training and testing sets in an 80:20 ratio to ensure proper performance evaluation.

The model performance was measured using accuracy, precision, recall, and F1-score. The CNN component effectively extracted hierarchical features, while the Capsule Network preserved spatial relationships between facial features, improving detection capability.

The hybrid CNN–Capsule model achieved an accuracy of 92.4%, precision of 91.2%, recall of 90.8%, and an F1-score of 91.0%. These results demonstrate improved robustness and generalization compared to individual models.

The training performance of the model is illustrated in Fig. 2, which shows the accuracy variation across training epochs. The model exhibits stable convergence with minimal overfitting.

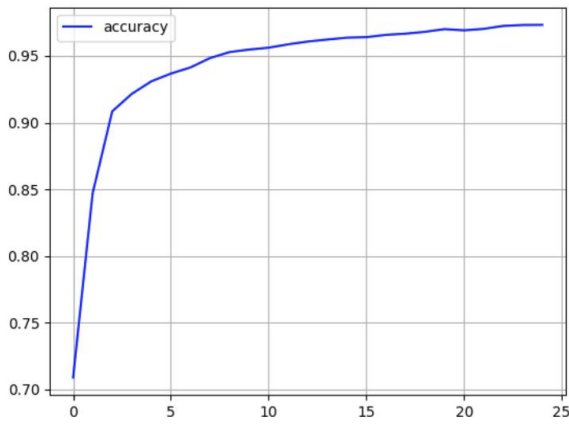


Fig. 2. Accuracy vs Epoch for the proposed CNN-Capsule model

The confusion matrix shown in Fig. 3 represents the classification performance of the model. Most of the real and fake samples are correctly classified, with only a small number of misclassifications.

|             | Real | Fake |
|-------------|------|------|
| Actual Real | 92   | 8    |
| Actual Fake | 7    | 93   |

Fig. 3. Confusion Matrix of deepfake detection results

Overall, the results confirm that the proposed hybrid approach provides reliable and efficient deepfake detection performance across different datasets.

## X. Conclusion

This study presented a deepfake detection approach based on Convolutional Neural Networks (CNN) and Capsule Networks. The proposed hybrid model combines effective feature extraction with spatial relationship modeling to improve detection performance.

Experimental results demonstrate that the hybrid CNN–Capsule model achieves high accuracy, precision, recall, and F1-score, outperforming individual models. The model shows strong generalization capability across different datasets and effectively identifies manipulated content.

Overall, the study highlights the importance of combining multiple deep learning techniques for reliable deepfake detection. The proposed approach provides a robust and efficient solution for real-world applications, while also offering scope for further improvements in scalability and computational efficiency.

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